Deliverable Guidelines

Quantitative Reasoning II · Final Project

This document provides detailed guidelines on final report writing and format to support your work. These are *guidelines*, not requirements. I expect your report to clearly communicate the information outlined in the project brief. Following these guidelines is one way to clearly communicate your work. I have also provided a project helper notebook that contains example code to produce regression tables of the format I've described in class, and to save your tables and figures to Google Drive.

1 Tone and style

Your report should be written in a professional, academic tone. Avoid casual language and slang. Use clear and concise sentences, and ensure that your arguments are logically structured. The report should guide the reader carefully through your research question, methodology, results, and conclusions. In more literary writing you might use all sorts of tools that do not apply here, such as building suspense, using synonyms, figurative language, or leaving room for interpretation. In this report, use the active voice where possible, refer to the same concept or variable by the same name throughout your report, and avoid vague statements. Practice saying exactly what you mean! Tell the reader what you did and show the reader evidence through your tables and figures.

2 Example report structure

Your report must be no more than 1,000 words of prose. My preference is that your prose comes first, uninterrupted by tables and figures. Then, present your tables and figures following your prose. Tables and figures should have as much space as they need to be readable (this might mean that a single figure takes up the whole page). Your prose, figures, and tables should all be contained in the same document. I suggest you render your report as a PDF to minimize chances that formatting issues arise when I download it from Populi. The structure and word count below is a guide — depending on your project, you may need to allocate your 1,000 word budget differently. The figures and tables you report do not count toward the word limit. You should only include tables and figures that you refer to in your prose. Each table and figure should be numbered and referred to by number in the prose. I expect reports to have somewhere between 1 and 3 tables and 1 and 3 figures, but the exact number will depend on your project and is up to your discretion.

Section	Contents	Word Count
Research question and background	What are you asking, and why should the reader care? Include here any vital background information for the reader to understand your motivation or design.	100
Data and design	Where does the data come from, and what analysis do you run?	350
Results	State your findings and their implications.	400
Limitations	What are the biggest limitations of your study?	150
		1,000

Research question and background: State what you are asking in one sentence. Clearly specify your X and your Y conceptually (*"I investigate..."*). Motivate your question and provide any background information the reader needs to understand your question. If you have a hypothesis, state it in this section.

Data and design: Orient the reader to your data and your research design. This might include:

- where the data comes from
- who collected the data
- how the data was collected
- what one row represents
- how many observations you have after any filters, merges, or transformations
- how your most important variables (X and Y) are measured
- what type of regression models you use (logistic? linear?)
- what type of control variables you include (you don't need to list out all of the variables – they should be apparent in your tables – describe the type and reason for inclusion).

You may, in this section, point the reader to your summary statistics, your correlation matrix, and any exploratory figure(s) that shaped what you did.

Results: In this section, state and interpret your key results. You don't need to interpret every coefficient on every regression you ran or report in your tables. Interpret the coefficient on your variable(s) of interest in terms of the units of your data (not just "significant" or "positive"). Describe what happens to the estimate(s) as you add controls and how you interpret these changes. If a control variable's coefficient surprised you or changed how you interpret your results, include that here.

Limitations: Every analysis has limitations! Briefly describe the biggest limitations (maybe 1-3). One well-explained limitation is better than a list of 10.

Remember: I'm not grading the significance of your results. It's okay if you find that your data cannot answer your question, or if the answer is not what you expected. The purpose of this project is to apply the tools we have learned in class, communicate your findings clearly, and demonstrate your understanding of the methods and their limitations.

3 Analysis suggestions

In case you're having a hard time charting your path through data analysis, this section reiterates some of the guidance I've given during class and individual student meetings.

3.1 Exploratory analysis

Before you run any regressions, explore your data. You should have a sense of the scale of your variables, how they relate to each other, and whether there are any patterns that might affect your analysis. The things you code here might not belong in your final report, but they are a valuable part of the research process.

After processing the raw data (filtering, merging, and creating new variables), you might do some or all of the following:

- **Summary statistics** for every variable you use. Summary statistics show you the scale and shape of the variables and whether any columns have missing data. Mean, median, and standard deviation of your variables are particularly useful when interpreting the magnitude of your regression coefficients later on.
- **A correlation matrix**, which shows the relationship between your variables and is helpful in designing your regression models and interpreting your results.
- **A histogram of your outcome**, which shows you the variation you are trying to explain.
- **A plot of your X against your Y:** is there a clear linear relationship? Are there outliers? Does the relationship look different depending on the value of a third variable?
- **Group-level comparisons** If you are interested in looking at how Y differs between groups, consider comparing the means of Y across the groups or plotting histograms or boxplots of Y for each group.

Choose the form of each figure from the kind of data you have (this is a replication of the data visualization table from the slide deck, with the python matplotlib commands added):

If you have	Use a	In matplotlib
Numerical y and numerical x	scatter plot	`plt.scatter(x, y)`
Numerical y and sequential x	line graph	`plt.plot(x, y)`
Numerical y alone	histogram	`plt.hist(y, bins=30)`
Numerical y and categorical x	boxplot	`df.boxplot(column="y", by="g")`
Summary statistics by group	bar plot	`df.groupby("g")["y"].mean().plot.bar()`

3.2 Building regression models

Your analysis should include at least one table of regression models. You should build your models progressively, the same way we did in class. This allows you to observe which variables affect your estimate, in what way. The following is an example of how this might look:

- Start with a simple regression of Y on X alone. This identifies a baseline estimate of the relationship you are interested in.
- Add control variables in whatever order makes the most intuitive sense. This may be one at a time, or may be in groups of similar variables.
- Put categorical variables and fixed effects in last.
- Report the progression in sequence.

You should have a coneptual reason for each model that you run. Don't run a model just because you haven't yet gotten the answer you want.

4 Formatting your tables and figures

4.1 Regression tables

Report similar models (models with the same dependent variable) together: one column per model, presented in the same order you built them. Here is an example. The project helper notebook contains code to produce a table of this format from your regression output:

Outcome: your Y, in its units	(1)	(2)	(3)
Your X	43.65	24.70	24.28
	(0.67)	(1.30)	(1.30)
A control		54.89	58.50
		(3.25)	(3.23)
R^2	0.33	0.35	0.36
Adjusted R^2	0.33	0.35	0.36
Some kind of fixed effect	No	No	Yes
n	8,812	8,812	8,812

Read across the first row: the estimate on X falls from 43.65 to 24.70 once the control enters column (2), and barely moves again in column (3). Discuss this type of movement in your Results section.

Some formatting best practices for your regression tables:

- Report the outcome and its units clearly at the top of the table.
- Standard errors of the coefficient in parentheses under each coefficient
- Fixed effects are not printed coefficient by coefficient. Indicate FEs with a Yes/No row says which models include them. *An exception to this best practice is if you have a small number of categories and you are interested in the coefficients.*
- R^2, adjusted R^2, and n at the foot of every column (if using logistic regression, you can substitute R^2 for pseudo-R^2).
- Represent your variables in the table with readable names rather than the column names in the dataset.

Produce regression tables with code and save to your Google Drive as an excel file or csv. Then, paste the table into your document as a table. Make sure each table is readable on its own without the prose. This means having a clear title, readable variable names, and a caption with a description of what the table reports.

4.2 Figures

Every figure should have:

- a title or caption describing what it shows
- axis labels, with units
- a legend, where appropriate

Keep in mind the plotting pitfalls we talked about in class! Save each figure from your code at a high enough resolution that it will be legible in your document. For example, in matplotlib you can save a figure with the following code:

```
plt.savefig("figure1.png", dpi=200, bbox_inches="tight")
plt.show()
```

4.3 Some final notes on the report that didn't fit elsewhere

- Use a readable and professional font and size (11 or 12 point). Use 1.5 or double spacing.
- Section headings don't count toward your word limit. You may use them to break up your prose into sections, but it's not required to do so.
- Put your word count at the top of the report
- Tables and Figures should be numbered. Your first Table should start at Table 1, and your first Figure should start at Figure 1. Refer to each table and figure by number in the prose, whether in the sentence or by parenthetical reference.